e-Hospital Clinical AI Platform

Comprehensive Project Status, Architecture Blueprint & Team Task Roadmap

Course: BMG5111 eHealth System Engineering	**Status: Production Core Ready / Deployment Phase**
Target Audience: Teammates, Technical Leads & Evaluators	**Date: October 2026 • Version: 2.4**

1. Executive Project Overview (For All Audiences)

What is this project? The **e-Hospital Clinical AI Platform** is an intelligent digital hospital assistant designed for Canadian primary care physicians and clinical staff. It solves one of healthcare's largest problems: **physician burnout caused by spending 2 to 3 hours every day on paperwork, patient communication, and documentation.**

The platform bridges natural doctor-patient interactions, automated remote patient monitoring via **WhatsApp**, hospital Electronic Health Records (**HL7 FHIR**), and automatic medical billing (**Ontario OHIP**) through a reliable artificial intelligence engine.

■ **Clinical AI Scribe** Listens to or reads doctor consultations, drafts complete clinical SOAP notes, and answers complex medical reasoning questions.	■ **WhatsApp Remote Care** Patients text blood pressure or heart rate from home; AI extracts readings and automatically updates their hospital chart.	■ **Ontario OHIP Billing** Automatically selects the correct OHIP billing fee codes (e.g. A007, K030) and ICD-10 diagnostic codes from the encounter.

2. How the System Works (Plain English Architecture)

The system consists of **5 coordinated layers** that work together seamlessly:

Layer	Component & Technology	What It Does (Simple Explanation)
1. User Interface	React 18, Vite, Mobile PWA	The visual web app used by doctors on mobile phones, tablets, or clinic desktops.
2. Central Server	FastAPI (Python) on Render Cloud	The brain of the platform. Handles requests, talks to the AI, and manages security 24/7.
3. AI Intelligence	OpenAI GPT-4o + Local Gemma AI	Dual-tier engine: uses cloud OpenAI for smart reasoning, with local offline AI as backup.
4. Patient Records	SQLite + AWS App Runner Cloud	Stores patient charts, appointments, vitals trends, and chat history permanently.
5. Messaging	Meta WhatsApp Cloud Gateway	Connects hospital doctors directly to patients' real smartphones via official WhatsApp.

3. Status Audit: What is 100% COMPLETED (Done)

The following core modules have been fully implemented, tested, and pushed to our central GitHub repository:

Module Name	Status	Key Features & Accomplishments

AI Clinical Scribe & Copilot	✔ DONE	• Voice dictation & audio recording in browser. • Converts freeform dialogue into structured SOAP clinical notes. • Mobile-friendly UI with visual keyboard lock so headers never shift. • Multi-session drawer: keeps separate, isolated chat histories per patient.
Database Chat Persistence (GPT-Style)	✔ DONE	• Central SQLite database table (chat_sessions) saves all conversations. • True multi-device synchronization: desktop chat immediately shows on mobile. • History persists permanently across page reloads and browser cache clears.
Dual-Tier Resilient AI Engine	✔ DONE	• Primary: OpenAI GPT-4o for high-level clinical reasoning. • Secondary: Local offline Ollama model (Gemma2:2b) with 50s timeout. • Tertiary: Verified deterministic clinical safety fallback (e.g. COPD inhaler guide).
WhatsApp Two-Way Patient Monitoring	✔ DONE	• Inbound webhook receives patient text messages. • Automated NLP vitals extractor parses BP/HR and updates EHR chart. • In-Portal Messenger allows doctors to text patients directly from clinic dashboard.
Cloud Deployment on Render	✔ DONE	• Deployed live at: bmg5111-ehealth-fall-2026.onrender.com. • Meta verification handshake tested and confirmed (HTTP 200 OK challenge). • Operates 24/7 without requiring personal laptop or terminal to be on.
FHIR R4 Patient Chart Management	✔ DONE	• Standardized healthcare interoperability format (HL7 FHIR R4). • Patient profiles, longitudinal blood pressure & heart rate graphing. • Integration with Central Clinical Cloud Network (AWS App Runner).
Medical Form Auto-Filling & Billing	✔ DONE	• OCR document scanner reads scanned PDF forms. • Auto-fills patient information onto official Ontario government forms. • Ontario OHIP billing code matcher (A007, K030, G310) with revenue tracker.

4. Action Roadmap: What is NOT DONE (Team Assignments)

To take this project from code-complete to a polished, award-winning academic submission, the remaining work has been broken down into **6 clear, bite-sized Task Packages** for teammates:

Task Package 1: Meta WhatsApp Live Activation & Credential Setup	Owner: Team Member 1 (Integration / Systems Lead) Difficulty: Moderate (No code required — portal configuration)

Objective: Connect our live Render server to the verified Meta WhatsApp Business Account.

Step-by-Step Action Items:
1. Log into Meta Business Suite & Developer Console (after verification).
2. Copy the Phone Number ID (15-digit numeric ID).
3. Generate a Permanent System User Token with messaging permissions.
4. Paste our Render Webhook Callback URL: https://bmg5111-ehealth-fall-2026.onrender.com/api/whatsapp/webhook.
5. Enter Verify Token: ehospital_verified_2026, click 'Verify and Save', and subscribe to 'messages'.
6. Add these keys into Render Environment Variables and test by sending a real WhatsApp message.

Task Package 2: OpenAI API Account Billing & Performance Benchmarking	Owner: Team Member 2 (AI & Clinical Evaluation Lead) Difficulty: Easy / Analytical

Objective: Reactivate GPT-4o cloud model and benchmark clinical accuracy against local AI.

Step-by-Step Action Items:
1. Add a small billing balance ($5 - $10) or valid key at platform.openai.com/billing.
2. Enter the key via the ■■ settings button in Scribe or in Render environment variables.
3. Benchmark: Run 5 standardized clinical cases (e.g. Robert Chen COPD, Sarah Khan Diabetes).
4. Compare GPT-4o vs local Gemma2:2b in terms of response time, clinical accuracy, and OHIP code precision.
5. Summarize findings into a 1-page evaluation table for the final report.

Task Package 3: Clinical Guidelines Knowledge Base (RAG Preparation)	Owner: Team Member 3 (Data & Medical Content Lead) Difficulty: Moderate (Data curation / Python)

Objective: Collect and format official Canadian clinical guidelines for AI grounding.

Step-by-Step Action Items:
1. Download official clinical practice PDF guidelines (Canadian Thoracic Society COPD, Diabetes Canada).
2. Organize documents in backend/data/guidelines folder.
3. Test Chart Chat endpoint (/api/chart-chat) with uploaded specialist consultation letters.
4. Verify that generated patient instructions cite specific Canadian guidelines.

Task Package 4: Physician Usability Testing & Mobile UI Polish	Owner: Team Member 4 (Frontend & User Experience Lead) Difficulty: Moderate (UI Testing & Feedback)

Objective: Validate mobile tablet experience and optimize clinical interface usability.

Step-by-Step Action Items:
1. Open https://bmg5111-ehealth-fall-2026.onrender.com on mobile phones (iPhone/Android) and iPad.
2. Test multi-session switching, history drawer opening, and keyboard dismissal on send.
3. Identify any layout quirks or small text sizes on mobile screens.
4. Verify speech synthesis voice feedback (text-to-speech) works clearly in a simulated noisy clinic.

Task Package 5: Healthcare Regulatory Compliance (PHIPA/PIPEDA) & Safety Audit	Owner: Team Member 5 (Health Policy & Quality Lead) Difficulty: Easy to Moderate (Documentation & Compliance)

Objective: Ensure project satisfies Canadian and Ontario healthcare privacy standards.

Step-by-Step Action Items:
1. Review Ontario PHIPA (Personal Health Information Protection Act) compliance criteria.
2. Document how our local offline fallback (Ollama) guarantees 100% on-premises patient data privacy.
3. Check audit log table in SQLite (activity_log) to ensure every physician access is logged.
4. Prepare a 2-page Privacy & Ethics Impact Assessment section for the final report.

Task Package 6: Video Walkthrough, Presentation Deck & Final Report	Owner: All Team Members / Project Lead Difficulty: Collaborative (Media & Communication)

Objective: Create a compelling 5-minute video presentation demonstrating the working system.

Step-by-Step Action Items:
1. Create a slide deck explaining: The Clinical Problem, System Architecture, Live Demo, and Results.
2. Record a live screen-capture video demonstrating:
a) Dictating a clinical visit in Scribe -> Generating SOAP note & OHIP billing.
b) Patient texting blood pressure on WhatsApp -> Chart automatically updating.
c) Mobile tablet view with synchronized chat history.
3. Compile project repository and documentation for course evaluator review.

5. Teammate Quickstart: How to Run the Project Locally

Any team member can run and explore the entire project on their own machine in 3 simple steps:

Step	Command / Action	Expected Output
1. Backend	cd backend python -m uvicorn main:app --reload	FastAPI server starts on http://localhost:8000. SQLite DB automatically boots up.
2. Frontend	cd frontend npm install npm run dev	React clinical dashboard opens on http://localhost:5173 with full interactive Scribe.
3. Cloud Demo	Open https://bmg5111-ehealth-fall-2026.onrender.com in any browser or phone	Live cloud version runs without running any local commands. Test on smartphones directly!

6. Project Delivery Milestones & Schedule

Milestone Phase	Target Date	Deliverable & Success Criteria	Status
Phase 1: Core EMR & AI Scribe	Completed	Clinical voice recording, SOAP note drafting, FHIR patient loader, database persistence.	✔ DONE
Phase 2: Cloud Deploy & WhatsApp	Completed	Deployed on Render, Meta verification webhook handshake, NLP vitals extraction.	✔ DONE
Phase 3: Live Meta Key & Testing	Upcoming (Week 1)	Insert live Meta WhatsApp token & phone ID, live test with real patient mobile numbers.	IN PROGRESS
Phase 4: Benchmarking & Guidelines	Upcoming (Week 2)	OpenAI GPT-4o quota top-up, Gemma vs GPT-4o accuracy comparison, guideline integration.	TO DO
Phase 5: Final Video & Submission	Final Milestone	5-minute demonstration video, final technical report, and project presentation.	TO DO

Summary & Next Actions for Team Members:
Please select your assigned Task Package from Section 4. If you have questions about running the code or need credentials, contact the Project Lead. All code updates should be committed to feature branches and merged into main following verification.